

Complexity and Noise-Resilience of Quantum Oracles for Symmetric Cryptanalysis

A Reproducible Study of Grover Key Search on a 4-bit SPN under Simulated NISQ Noise

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September 2, 2026

Abstract

We construct and validate a reversible quantum oracle for known-plaintext key search against a toy 4-bit substitution-permutation network, apply Grover’s algorithm [15] to recover the key, and quantify how the attack degrades under simulated NISQ noise [29]. The oracle evaluates the cipher on the key register in superposition and compares the result against known ciphertexts, so at no point does it presuppose the secret key. Its correctness is established exhaustively: the encryption circuit’s exact unitary is verified against an independent classical reference for every key and plaintext, and the oracle is shown to restore its data registers completely, which is the property on which Grover interference depends.

Ideal simulation reproduces the closed-form amplification curve $\sin^2((2k+1)\theta)$ to within 4.33×10^{-15} , including the over-rotation decline beyond the optimal iteration count. Under depolarizing and readout noise at contemporary hardware rates ($p_2 = 10^{-2}$), the 1,404-CX circuit is driven to an output entropy of 3.995 bits against a maximum of 4.000: amplification is not merely degraded but absent. We then apply classical readout-error mitigation by assignment-matrix inversion and obtain the paper’s central result, a *boundary*. When measurement error is the only channel, mitigation restores the success probability from 0.8408 to 0.9652 against an ideal 0.9613; when gate error is also present it moves the result only from 0.0574 to 0.0596, both of which are random guessing. We further show that unconstrained pseudo-inverse mitigation returns a negative probability mass of 0.0049 and *overshoots* the ideal value, whereas a non-negative least-squares formulation remains physical. A noise sweep places the largest two-qubit error rate still permitting recovery at 3.0×10^{-3} , one to two orders of magnitude below current devices, and a resource scan shows the quadratic query saving being purchased with circuits growing from 1,404 to 33,360 CX gates between 4- and 8-bit keys. All code, data and figures are released for reproduction.

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1 Introduction

Grover’s algorithm provides a quadratic speedup for unstructured search [15] and this bound is known to be optimal [5, 32]. Applied to symmetric cryptography, it reduces the cost of exhaustive key search from $\mathcal{O}(2^n)$ to $\mathcal{O}(2^{n/2})$ oracle queries, which is the standard justification for doubling symmetric key lengths in post-quantum guidance [6, 12].

That query-complexity statement, however, says nothing about whether such an attack is *implementable*. Two questions sit between the asymptotic speedup and a real threat:

1. **What does one oracle query actually cost?** The oracle must contain a full reversible implementation of the cipher. Detailed resource estimates for AES [14, 17] and SHA [2] show this circuit, not the query count, dominates the cost.
2. **Does amplification survive hardware noise?** Grover depends on coherent interference across the whole search register. Present-day devices are noisy [29], and error mitigation, while able to extend useful computational reach [11, 19], is not error correction.

This report addresses both questions on a deliberately small but genuine instance. We use a 4-bit substitution–permutation network built from the published PRESENT S-box [7] in the structure of the classical SPN teaching model [16]. The key space is tiny, which is precisely what makes exhaustive verification of every claim affordable: we check the quantum circuit against a classical reference for *all* keys and plaintexts rather than sampling.

1.1 Contributions

- A **non-circular** reversible key-search oracle: it evaluates $E_K(P)$ in superposition and compares against known ciphertexts, rather than being handed the target key. It uses $n + 4r$ qubits and **no ancillas**.
- **Exhaustive correctness verification** of both the encryption unitary and the oracle’s uncomputation, the latter being a failure mode that produces correct-looking phases with silently broken amplification.
- A measured demonstration that **a single known-plaintext pair does not determine the key**, reproducing at toy scale the multi-block requirement of the AES oracle literature [14, 17], together with the gap between the information-theoretic bound and the number of pairs actually needed.
- **Localisation of the boundary of readout-error mitigation**: near complete recovery of a readout-limited signal, and no recovery whatsoever of a gate-error-limited one, measured under three noise models that differ only in gate error.
- A demonstration that **unconstrained inversion is unphysical** in practice, not just in principle, reproducing the unfolding pathology [26] at small scale.
- An explicit account of the **simulation resource trade**: trajectory sampling reduces noisy-simulation memory from $\mathcal{O}(4^n)$ to $\mathcal{O}(2^n)$, converting a memory barrier into a runtime cost.

2 Background

2.1 Symmetric key search as an oracle problem

In a known-plaintext attack the adversary holds pairs (P_i, C_i) with $C_i = E_K[K^*](P_i)$ for an unknown key K^* , and seeks

$$K^* \in \left\{ K \in \{0, 1\}^n : E_K(P_i) = C_i \ \forall i \right\}. \quad (1)$$

Classically this requires up to 2^n trial encryptions. The corresponding Boolean predicate

$$f(K) = \begin{cases} 1 & \text{if } E_K(P_i) = C_i \text{ for every } i, \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

is exactly the marking function Grover’s algorithm requires. Note that Eq. (2) references only (P_i, C_i) — the data an attacker possesses — and never K^* .

2.2 Reversibility

Quantum evolution is unitary, hence reversible, so an irreversible classical function cannot be inserted into a circuit directly. The standard remedy embeds it as

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle, \quad (3)$$

a construction resting on Bennett’s demonstration that any computation can be made logically reversible [4], itself motivated by Landauer’s result that irreversible bit erasure carries an unavoidable thermodynamic cost [22]. For a *phase* oracle we instead want

$$O_f |K\rangle = (-1)^{f(K)} |K\rangle, \quad (4)$$

which Section 4 realises without any ancilla register.

2.3 Grover search

Beginning from the uniform superposition $|s\rangle = N^{-1/2} \sum_{x=0}^{N-1} |x\rangle$ with $N = 2^n$, Grover alternates the phase oracle (4) with the diffusion operator $D = 2|s\rangle\langle s| - I$. With M marked states and $\theta = \arcsin \sqrt{M/N}$, the probability of measuring a marked state after k iterations is

$$P(k) = \sin^2((2k+1)\theta), \quad (5)$$

maximised at

$$k^* = \left\lfloor \frac{\pi}{4} \sqrt{\frac{N}{M}} \right\rfloor. \quad (6)$$

Two consequences of Eq. (5) are frequently misstated and are measured directly in Section 6.1. First, k^* is *not* $\sqrt{N}$; the factor $\pi/4$ and the dependence on M both matter [8]. Second, $P(k)$ is periodic, so exceeding k^* *reduces* the success probability. Amplitude amplification generalises this to arbitrary state preparation [9]; a full derivation appears in Nielsen and Chuang [28, Ch. 6].

2.4 NISQ noise and error mitigation

Contemporary processors operate without error correction, in what Preskill [29] termed the NISQ regime. We model two channels. The depolarizing channel replaces a state with the maximally mixed state with probability p ,

$$\mathcal{E}_p(\rho) = (1-p)\rho + p \frac{I}{2^m}, \quad (7)$$

applied at rate p_1 to single-qubit and p_2 to two-qubit gates. Readout error is an asymmetric classical bit flip at measurement time.

Error *mitigation* differs fundamentally from error correction: it applies classical post-processing to recover expectation values or distributions without correcting any quantum state [11]. Two families are relevant. Zero-noise extrapolation runs a circuit at deliberately amplified noise and extrapolates to the zero-noise limit [19, 24, 31]. Readout-error mitigation, which we implement, characterises the measurement channel and inverts it [10, 27]. The distinction matters for interpreting our results: readout mitigation addresses only the measurement step, and Section 6.3 measures the consequence.

3 The cipher

We use a 4-bit block SPN with an n -bit master key ($n = 4$ by default) and r rounds ($r = 2$):

$$E_K(P) = \text{ARK}_{rk_r} \circ (\pi \circ S) \circ \cdots \circ \text{ARK}_{rk_1} \circ (\pi \circ S) \circ \text{ARK}_{rk_0}(P), \quad (8)$$

giving r substitution layers and $r+1$ key additions.

Substitution. S is the PRESENT 4-bit S-box [7], a published cryptographic primitive rather than an ad-hoc permutation:

x	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
$S[x]$	C	5	6	B	9	0	A	D	3	E	F	8	4	7	1	2

Permutation. π is the fixed bit permutation ($0 \mapsto 1, 1 \mapsto 3, 2 \mapsto 0, 3 \mapsto 2$), deliberately not a rotation so that diffusion is not trivially self-similar with the key schedule.

Key schedule. Round key $rk_j = \text{rotl}(K, j) \wedge 0\text{x}\text{F}$. This choice is made for a quantum-implementation reason: a rotation is *free* on a quantum register, since XOR-ing $\text{rotl}(K, j)$ into the data register only changes *which key qubit acts as the CNOT control*. A schedule involving substitutions would have to be computed and uncomputed inside every oracle call.

The classical implementation (`spn.py`) depends on nothing beyond the Python standard library and shares no code with the quantum implementation. This independence is deliberate: it makes the equivalence check of Section 5.5 a genuine cross-check rather than a tautology.

4 The reversible oracle

4.1 Construction

The oracle is built as **compute – phase – uncompute**. Let U_E be the reversible, key-controlled encryption acting in place on a 4-qubit data register, and let there be r known pairs, each with its own data register initialised to $|P_i\rangle$.

1. **Compute.** Apply U_E to each data register, with the key register providing CNOT controls only:

$$\sum_K |K\rangle |P_1 \cdots P_r\rangle \mapsto \sum_K |K\rangle |E_K(P_1) \cdots E_K(P_r)\rangle.$$

2. **Phase.** Conjugate a multi-controlled Z over all $4r$ data qubits by an X -mask encoding the concatenated target ciphertext, so a -1 phase is applied exactly when every register holds its target.
3. **Uncompute.** Apply $U_E^\dagger$ to each data register.

Because the phase is diagonal it commutes through step 3, giving

$$\sum_K |K\rangle |P_1 \cdots P_r\rangle \mapsto \sum_K (-1)^{f(K)} |K\rangle |P_1 \cdots P_r\rangle, \quad (9)$$

with the data registers restored and therefore factored out.

4.2 Why uncomputation is decisive

This is the construction’s most important subtlety. Omitting step 3 still yields an oracle that marks the correct states with the correct phases. But the key register is then entangled with the data registers, and that residual which-path information destroys the interference on which the diffusion operator relies: amplification silently fails to accumulate. It is a defect that inspects as correct and behaves as wrong, which is why Section 5.5 tests for it explicitly, asserting that the entire probability mass returns to the restored-plaintext subspace.

4.3 Ancilla-free by construction

The oracle uses $n + 4r$ qubits and no ancillas. Two structural facts make this possible. Every cipher layer — key XOR, S-box, bit permutation — is individually a bijection on 4 bits and therefore acts in place. And the r separate 4-bit equality tests are fused into a *single* $4r$ -qubit AND condition, rather than computing r flag qubits and combining them; flags would themselves require uncomputation.

Table 1: Consistent keys versus number of known plaintext/ciphertext pairs, measured exhaustively over all 16 keys ($n = 4$, $r = 2$).

Known pairs	Worst-case M	Mean M	Keys uniquely determined
1	3	1.88	8 / 16
2	2	1.25	12 / 16
3	1	1.00	16 / 16

4.4 One plaintext pair is not enough

For a 4-bit block, $K \mapsto E_K(P)$ is not injective, so a single known pair generally leaves several consistent keys. Exhaustive measurement over the whole key space at $r = 2$ gives Table 1.

This is not an artefact of the toy scale. It is the same counting argument that requires quantum key-search oracles against AES-128 to encrypt several plaintext blocks [14, 17]. Measured across key widths, the information-theoretic estimate $\lceil n/4 \rceil + 1$ is a *lower* bound: at $n = 4$ between 1 and 3 pairs are needed depending on the key, at $n = 6$ exactly 3, and at $n = 8$ between 3 and 4. The implementation therefore grows the pair count until classical brute force confirms $M = 1$, and reports the number used. Each additional pair costs a further 4-qubit register and two more encryption circuits per oracle call, making this the dominant cost driver.

5 Method

5.1 Simulation and the memory–time trade

Simulation uses Qiskit [18] with the Aer backend. One configuration choice dominates feasibility. Noisy simulation admits two representations, and their scaling differs qualitatively:

Table 2: Memory required for noisy simulation. Aer’s `automatic` method may select density-matrix simulation once a noise model is attached.

Method	Scaling	8 qubits	16 qubits	20 qubits
Density matrix	$\mathcal{O}(4^n)$	1 MB	68 GB	17.6 TB
Statevector trajectories	$\mathcal{O}(2^n)$	4 KB	1 MB	16 MB

We therefore pin the simulation method to `statevector`, which makes Aer use stochastic quantum trajectories: one pure state per shot, sampling a Kraus operator at each noisy gate, with the shot average converging to the same density-matrix result. This reduces memory from $\mathcal{O}(4^n)$ to $\mathcal{O}(2^n)$ at the cost of runtime scaling as

$$T \propto \text{shots} \times \text{gates} \times 2^{n_{\text{qubits}}}. \quad (10)$$

It is worth stating precisely what is and is not saved here. Shot-based sampling does not avoid storing a statevector — Aer maintains one either way. What trajectory sampling avoids is the *density matrix*. The saving is $4^n \rightarrow 2^n$, and it is what allows this study to run on a single laptop.

A transpilation prerequisite. A noise model binds errors to *named basis gates*. A circuit still containing composite instructions (a generic `UnitaryGate`, a multi-controlled Z , a `swap`) contains no `cx` or `sx` for the model to attach to, and would execute essentially noise-free while appearing to be noisy — with no error raised anywhere. Every execution path therefore transpiles to the IBM-native basis $\{\text{id}, \text{rz}, \text{sx}, \text{x}, \text{cx}\}$ first. The S-box is synthesised from its permutation matrix by quantum Shannon decomposition [30]. `rz` is left noiseless: on transmon hardware a Z

rotation is a virtual frame change of zero duration, applied by shifting the phase of subsequent drive pulses rather than by playing a pulse [25], and IBM’s backend properties report no gate error for it. Section 8 records the sense in which this is an idealisation.

5.2 Noise parameters

Single-qubit depolarizing rate $p_1 = 10^{-3}$, two-qubit rate $p_2 = 10^{-2}$, and asymmetric readout error with $P(1|0) = 2\%$ and $P(0|1) = 4\%$. The order-of-magnitude gap between p_1 and p_2 , and the readout asymmetry (energy relaxation makes $1 \rightarrow 0$ more likely), both follow measured superconducting-device behaviour.

5.3 Readout-error mitigation

Measured and true distributions are related through the assignment matrix,

$$y = Ax, \quad A_{ij} = P(\text{observe } i \mid \text{prepared } j), \quad (11)$$

characterised by 2^n calibration circuits over the measured qubits. Only the key register is measured, which keeps calibration at 16 circuits rather than 2^{n+4r} . Calibration circuits are shallow, so A captures the measurement channel rather than a mixture of gate and readout effects.

We implement three inversions of $y = Ax$:

pinv the Moore–Penrose pseudo-inverse. Unbiased, but A^{-1} is not a stochastic matrix, so the result may leave the probability simplex [26].

clip pinv followed by clipping and renormalisation: cheap, but the clip biases the estimate.

nnls $\min_x \|Ax - y\|_2$ subject to $x \geq 0$, then renormalised. Always a valid distribution, which is why it is our default. Two caveats, so the attribution is not overclaimed. This is not the mechanism used by M3 [27], which is matrix-free – never forming A or its inverse – and solves within the subspace spanned by the observed bitstrings; M3 shares the goal of avoiding exponential calibration, not the method. And Nachman et al. [26], cited above against **pinv**, in fact advocate iterative Bayesian unfolding over both matrix inversion *and* least squares. We retain NNLS because it is exactly solvable and sufficient to establish the readout-versus-gate boundary measured in Section 6.3, not because it is optimal.

We also report $\text{cond}(A)$, the amplification factor for statistical noise under inversion — the reason mitigation is not free.

5.4 Experimental design

A — Ideal. Sweep k with no noise; compare against Eq. (5). Establishes oracle correctness and the over-rotation effect.

B — Noisy. Repeat under depolarizing and readout noise.

C — Mitigation. Apply all three inversions under *three* noise models that hold readout error fixed and vary only gate error: readout-only, reduced gate error ($\times 0.03$), and full gate error. Any difference between them is therefore attributable to gate noise. Running only the first would overstate the technique; only the last would understate it.

D — Threshold. Scale all error rates and locate where recovery fails, defined operationally as the correct key ceasing to be the single most likely outcome.

E — Resources. Transpiled cost and query complexity versus key width. Circuits here are transpiled but *not* simulated: by Eq. (10) simulating the largest is infeasible, whereas exact resource counts cost seconds.

All runs use 4,096 shots and seed 20260902 (Qiskit 2.5.2, Aer 0.17.2, Python 3.13.3). Every reported probability carries a binomial standard error, and comparisons against the random-guessing baseline require a two-sigma margin before an advantage is claimed.

5.5 Validation

Two exhaustive checks underpin every downstream result.

The circuit is the cipher. For every key and every plaintext, across five configurations, the *exact unitary* of the encryption circuit is compared against the independent classical reference:

$$U_E |K\rangle |X\rangle = |K\rangle |E_K(X)\rangle \quad \forall K, X. \quad (12)$$

This tests the circuit itself, not a sampled outcome.

The oracle uncomputes. The oracle is checked to apply -1 to exactly the marked keys and $+1$ elsewhere, *and* to return the full probability mass to the restored-plaintext subspace,

$$\sum_K |\langle K, P_1 \cdots P_r | O_f | \psi \rangle|^2 = 1, \quad (13)$$

which is the assertion that catches incomplete uncomputation.

The oracle is additionally verified to be an involution ($O_f^2 = I$), and the mitigation code is checked against an analytically constructed assignment matrix so that recovery is verified *exactly* rather than merely shown to improve.

6 Results

6.1 Experiment A: the implementation is exact

For the instance $n = 4$, $r = 2$, secret key 1101 with 1 known pair and $M = 1$, Eq. (6) gives $k^* = 3$. Simulation yields a peak success probability of 0.9613 at $k = 3$, and agrees with Eq. (5) to a maximum deviation of 4.33×10^{-15} (Figure 1).

That deviation is at the level of floating-point rounding. The implementation is not approximately Grover; within double precision it is Grover. This is the result that licenses interpreting everything downstream as a statement about noise rather than about a bug.

Two features of Eq. (5) are visible directly. The peak is 0.9613, not unity: with $N = 16$ and $M = 1$, $(2k + 1)\theta$ never lands exactly on $\pi/2$ for integer k , so certainty is unreachable. And over-rotation is confirmed (yes): success falls from 0.9613 at $k = 3$ to 0.5817 at $k = 4$, discarding roughly a third of the success probability through one extra iteration.

6.2 Experiment B: contemporary error rates destroy the attack

At $p_1 = 10^{-3}$ and $p_2 = 10^{-2}$ the transpiled search circuit contains 1,404 CX gates at depth 4,861. Table 3 and Figure 2 show the outcome.

At k^* the noisy success probability is 0.0574 ± 0.0036 against an ideal 0.9613, with TVD = 0.9039 and an output entropy of 3.995 bits against a maximum of 4.000. The output is statistically indistinguishable from uniform, and the measured advantage over random guessing is -1.4 standard errors (significant: no).

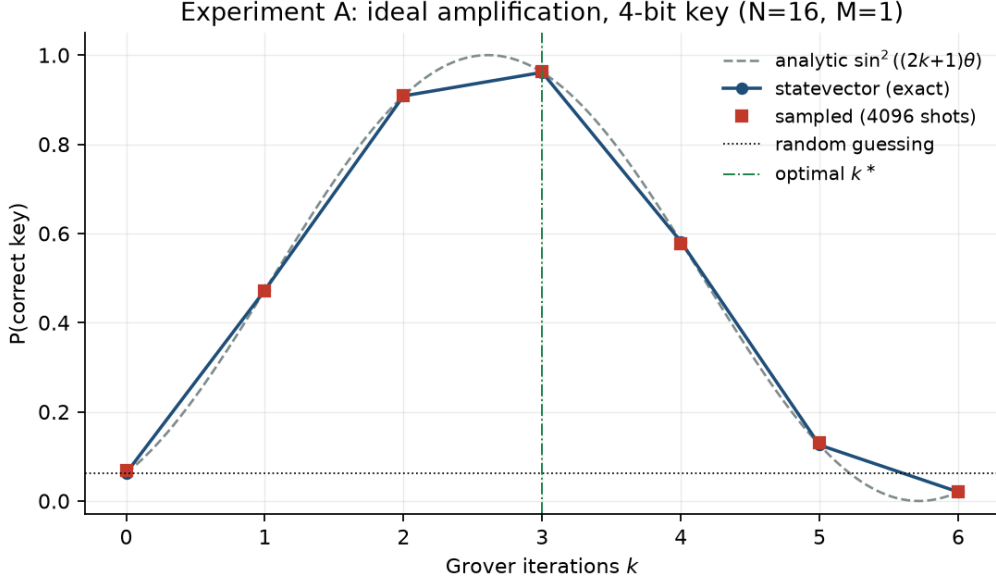


Figure 1: Experiment A. Exact statevector probability (circles) and sampled result (squares) against the closed form $\sin^2((2k+1)\theta)$ (dashed). The decline beyond k^* is the over-rotation predicted by Eq. (5), not a numerical artefact.

Table 3: Experiment B. Ideal versus noisy success probability across the iteration sweep; k^* in bold. Entropy is in bits, against a uniform maximum of 4.000.

k	CX	Depth	Ideal P	Noisy P	TVD	Entropy
0	0	4	0.0625	0.0593	0.026	3.997
1	468	1,623	0.4727	0.0647	0.408	3.996
2	936	3,242	0.9084	0.0583	0.850	3.996
3	1,404	4,861	0.9613	0.0574	0.904	3.995
4	1,872	6,480	0.5817	0.0618	0.520	3.997
5	2,340	8,099	0.1255	0.0557	0.073	3.996
6	2,808	9,718	0.0204	0.0623	0.056	3.996

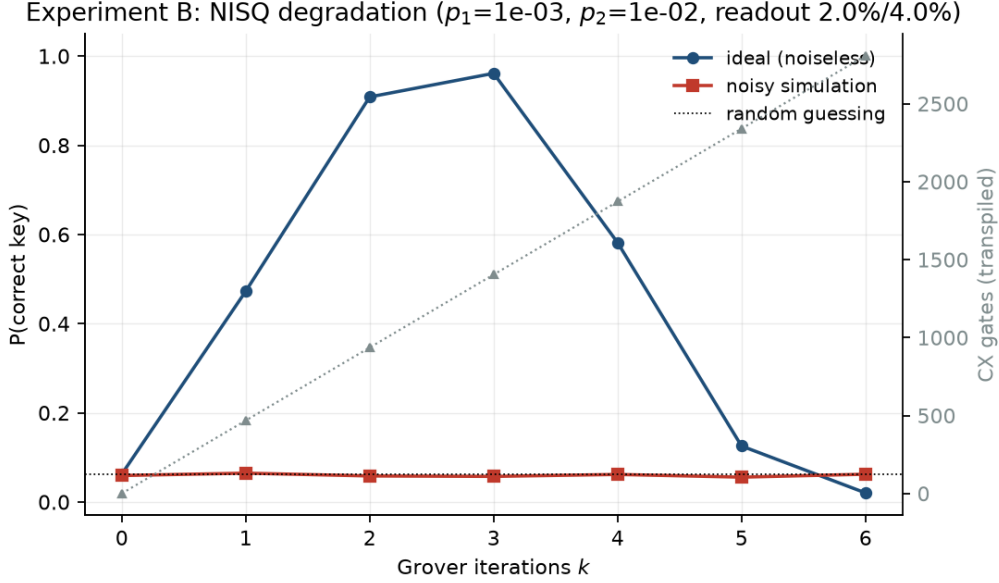


Figure 2: Experiment B. The noisy curve is flat at $1/N$ irrespective of iteration count: amplification does not occur at all. The dotted trace shows the transpiled CX count growing linearly with k .

The correct reading is stronger than “degraded”. The noisy curve does not track a reduced version of the ideal curve; it is flat, at the random-guessing level, for every iteration count. Amplification is absent rather than diminished. This is consistent with the conclusion of Jaques et al. [17] that Grover-based key search lies far beyond near-term hardware, and we report it without tuning the noise down to obtain a more encouraging figure.

6.3 Experiment C: the boundary of readout mitigation

This is the report’s central result. Table 4 gives all three noise models and all three inversions; Figure 3 shows the distributions.

Table 4: Experiment C. Readout error is held fixed at $P(1|0) = 2\%$, $P(0|1) = 4\%$ in all three scenarios; only gate error varies. Ideal $P = 0.9613$. “Valid” means the result is a probability distribution.

Scenario	Method	$P(\text{key})$	TVD	Neg. mass	Valid
Readout only	raw	0.8408	0.1225	—	yes
	pinv	0.9692	0.0143	0.0049	no
	clip	0.9644	0.0119	0.0000	yes
	nnls	0.9652	0.0127	0.0000	yes
+ reduced gate ($\times 0.03$)	raw	0.6343	0.3270	—	yes
	pinv	0.7325	0.2288	0.0000	yes
	clip	0.7325	0.2288	0.0000	yes
	nnls	0.7325	0.2288	0.0000	yes
+ full gate error	raw	0.0574	0.9039	—	yes
	pinv	0.0596	0.9017	0.0000	yes
	clip	0.0596	0.9017	0.0000	yes
	nnls	0.0596	0.9017	0.0000	yes

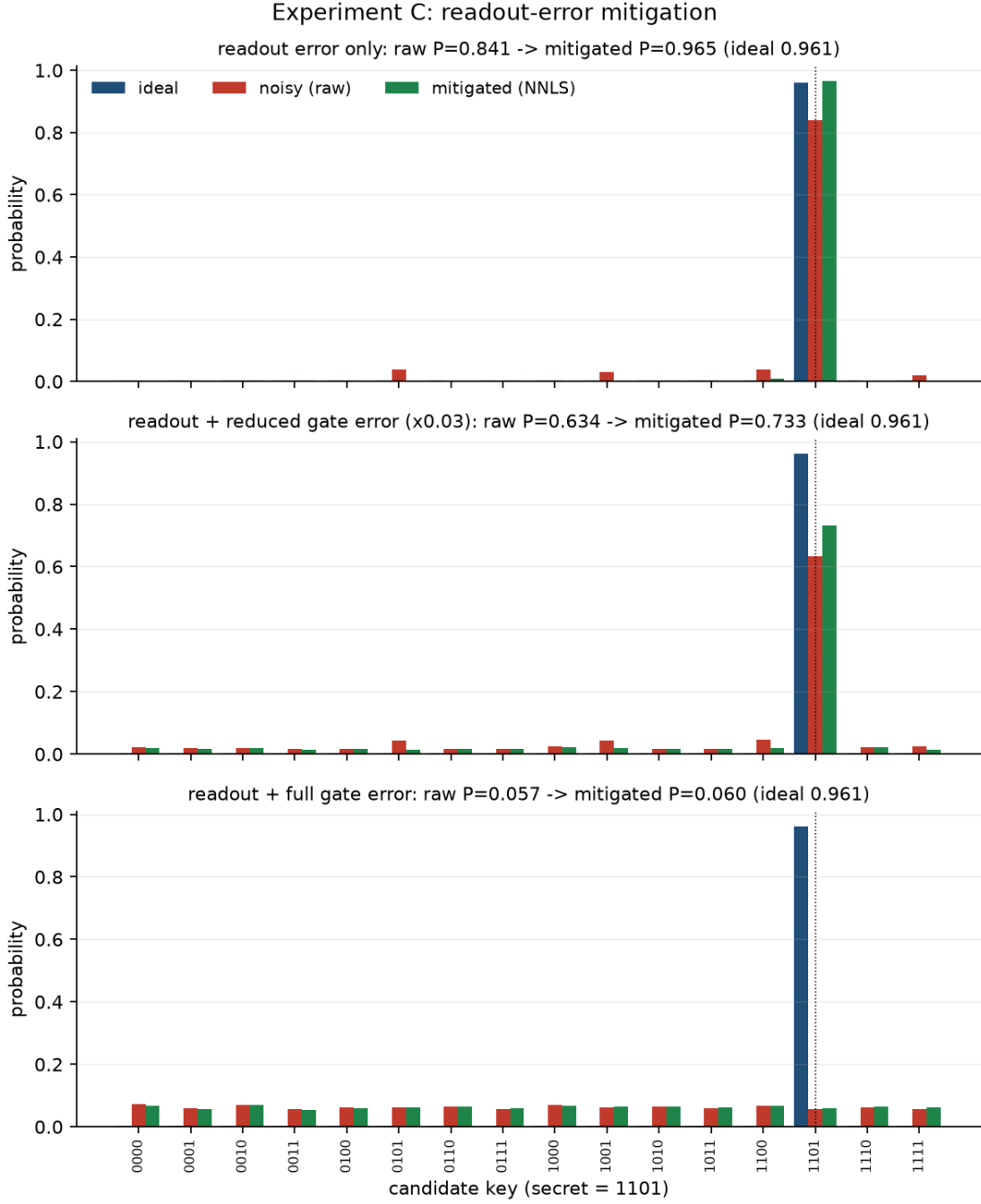


Figure 3: Experiment C. Per-key distributions for the three noise models. Mitigation restores the peak when readout error is the only channel (top) and cannot restore it once gate error dominates (bottom).

Readout-limited: near-complete recovery. With measurement error as the only channel, mitigation raises the success probability from 0.8408 to 0.9652 against an ideal 0.9613, and reduces the total variation distance from 0.1225 to 0.0127. Since A fully describes the corruption, inversion recovers the ideal distribution to within shot noise. Mean readout fidelity is 0.8852 with $\text{cond}(A) = 1.30$.

Gate-limited: no recovery at all. With gate error at the modelled rate, mitigation moves the result from 0.0574 to 0.0596. Both are random guessing ($1/N = 0.0625$). Nothing is recovered, and this is the correct outcome: readout mitigation is post-processing on a classical histogram, and once gate error has destroyed the quantum state, the information is gone before measurement occurs. No classical operation on the histogram can restore it.

The unphysicality of unconstrained inversion. In the readout-only scenario, `pinv` reports $P = 0.9692$ — *higher* than the ideal 0.9613 — while placing 0.0049 of probability mass below zero, so it is not a valid distribution (valid: no). This is not a better answer but an unphysical overshoot, reproducing at small scale the unfolding pathology documented by Nachman et al. [26]. The non-negative least-squares formulation sacrifices a little accuracy to remain inside the probability simplex, which is the correct trade and the reason it is our default.

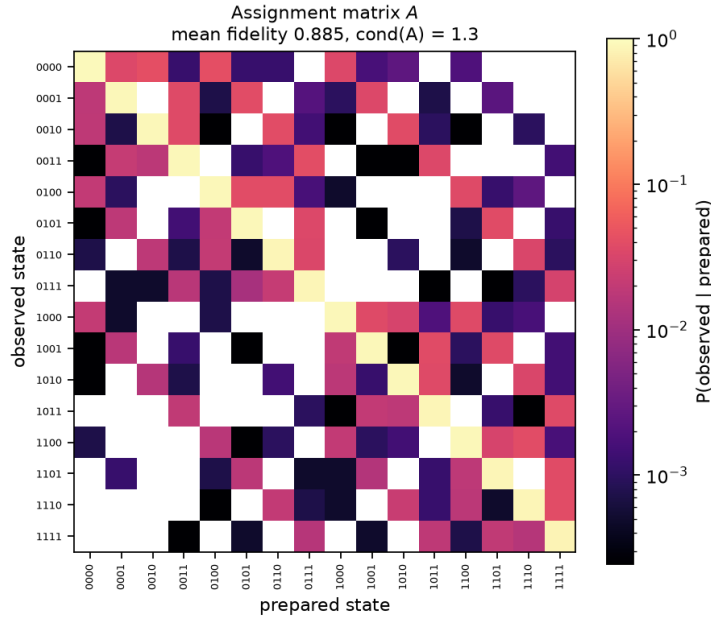


Figure 4: The characterised assignment matrix A for the measured key register, on a logarithmic colour scale. Off-diagonal structure is the readout error that mitigation inverts.

6.4 Experiment D: how much better must hardware be?

The largest two-qubit error rate at which the correct key remains the most likely outcome is 3.0×10^{-3} ; by 10^{-2} the attack has failed. Since contemporary superconducting two-qubit error rates sit in the 10^{-2} – 10^{-3} range, this circuit requires roughly one to two orders of magnitude improvement.

The qualifier matters more than the number: this is a 16-key search. The requirement is not that physical gate fidelities improve somewhat, but that cryptographically relevant instances demand error correction [1, 13] rather than better physical gates alone.

Table 5: Experiment D. Success against two-qubit error rate for the 1,404-CX, depth-4,861 circuit. “Checks” is the expected number of classical trial encryptions if candidates are verified in descending probability order.

p_2	$P(\text{key})$	Rank of key	Checks	Key ranked first
0 (ideal)	0.9636	1	1.0	yes
10^{-5}	0.9495	1	1.0	yes
3.0×10^{-5}	0.9358	1	1.0	yes
10^{-4}	0.8687	1	1.0	yes
3.0×10^{-4}	0.7300	1	1.0	yes
10^{-3}	0.4048	1	1.0	yes
3.0×10^{-3}	0.1282	1	1.0	yes
10^{-2}	0.0574	13	13.0	no
3.0×10^{-2}	0.0530	14	14.0	no

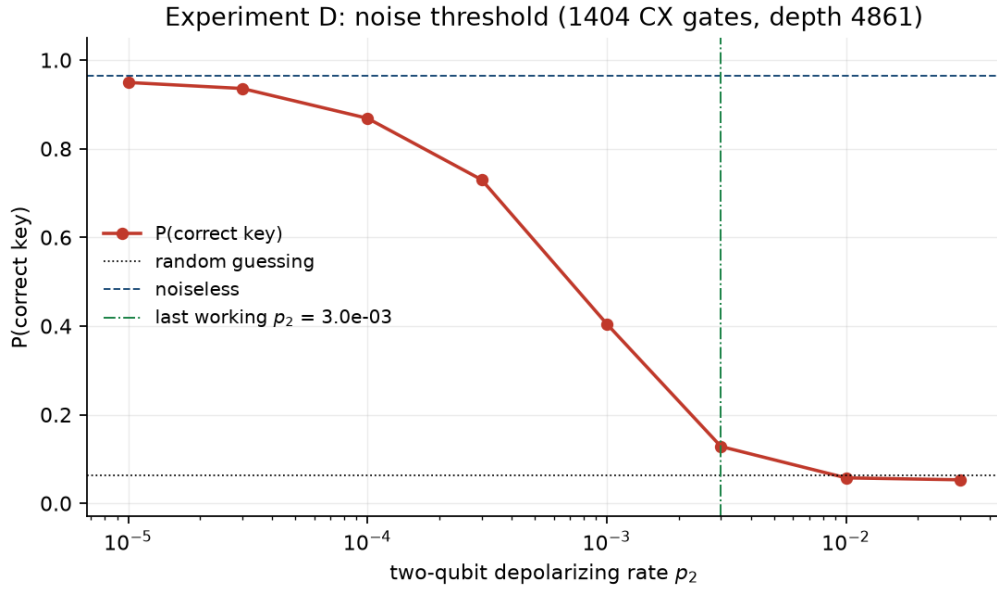


Figure 5: Experiment D. Success probability against two-qubit depolarizing rate. The collapse is sharp because error accumulates over 1,404 CX gates.

6.5 Experiment E: the speedup is in queries, and queries are not free

Table 6: Experiment E. Transpiled resources and query complexity against key width. Circuits are transpiled but not simulated. “Speedup” is $2^n/k^*$, the query-count ratio against classical exhaustive search.

n	N	Pairs	Qubits	k^*	CX	Depth	Query speedup
4	16	1	8	3	1,404	4,861	$5.3\times$
5	32	2	13	4	3,824	13,212	$8.0\times$
6	64	2	14	6	5,784	19,924	$10.7\times$
7	128	2	15	8	11,232	39,172	$16.0\times$
8	256	3	20	12	33,360	96,604	$21.3\times$

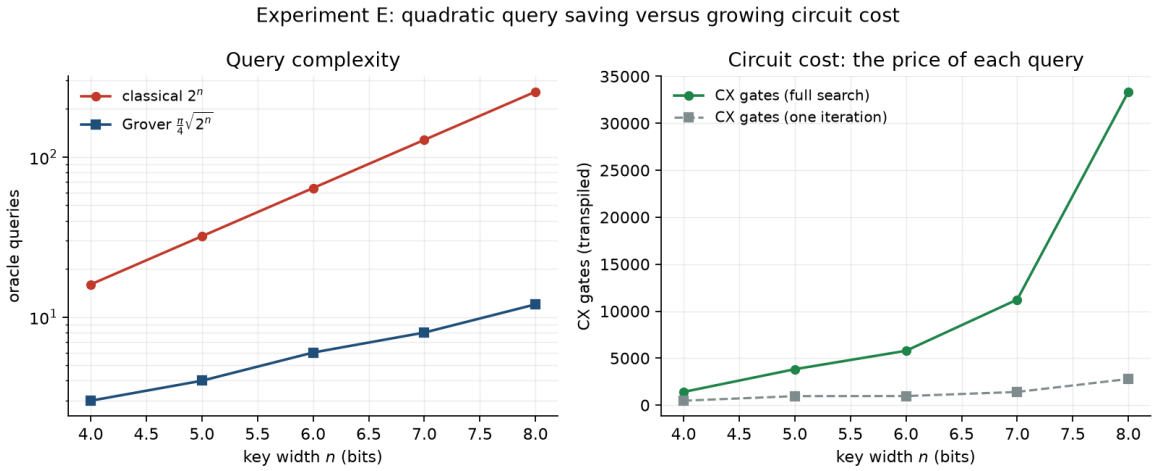


Figure 6: Experiment E. The quadratic query saving (left) set against the circuit cost of each query (right).

The quadratic saving is real and visible in the k^* column. But each query costs a circuit, and the CX count rises from 1,404 at $n = 4$ to 33,360 at $n = 8$ — driven jointly by the cipher, the iteration count, and the number of plaintext pairs required for a unique key (Section 4.4).

The 8-bit configuration also illustrates Section 5.1 concretely: at 20 qubits, density-matrix simulation would need 16384.0 GB, whereas a trajectory statevector needs 16.000 MB. The memory barrier is removed; by Eq. (10) the runtime cost remains, which is why these circuits are counted rather than simulated.

This distinction — between a query-complexity speedup and a resource estimate — is the connection to the AES literature [14, 17]: the query count is the easy part.

The speedup does not parallelise. One further caveat belongs with any statement of the quadratic saving, because it is what makes the saving far less threatening in practice. Grover’s algorithm *cannot be parallelised better than by partitioning the search space across independent quantum computers* [32]. Splitting a search of size N across p machines gives each a subspace of size N/p and hence $\sqrt{N/p}$ queries, a saving of only $\sqrt{p}$ rather than the factor p that classical exhaustive search enjoys. The $\Omega(\sqrt{N})$ lower bound [5, 32] means no better schedule exists.

The consequence is that the k^* column of Table 6 is essentially a *sequential depth*, not a cost that can be spent down with more hardware. Combined with the CX counts beside it, this is why Jaques et al. [17] conclude that Grover key search against AES remains out of reach even

under generous assumptions: the attack is depth-bound, and depth is precisely what a noisy device cannot supply. Our Experiment D threshold is the same conclusion in miniature.

7 Discussion

Taken together the experiments give a consistent and unflattering picture of Grover-based cryptanalysis on unmitigated NISQ hardware.

The algorithm is not the bottleneck. Experiment A shows the oracle and amplification are exact to floating-point precision. Nothing in the algorithm limits the attack; what limits it is the physical cost of executing 1,404 two-qubit gates coherently.

Mitigation is channel-specific, and knowing which channel matters. The natural expectation is that error mitigation partially recovers a degraded signal. Experiment C shows the truth is sharper and more useful: recovery is near-total when the mitigated channel is the dominant one and exactly nil otherwise. Reporting an average over both regimes would have obscured this. The practical consequence is that one should characterise *which* channel dominates before selecting a mitigation strategy — and for deep circuits the answer will be gate error, for which zero-noise extrapolation [24, 31] rather than readout inversion is the appropriate tool.

Unphysical estimators are a real hazard. That `pinv` reported a success probability above the ideal value, with negative probability mass, is worth emphasising because the failure is not loud. It produces a plausible number that happens to be impossible. Constrained inversion should be the default, and validity should be asserted rather than assumed.

Oracle design costs more than oracle queries. The requirement for multiple plaintext pairs (Section 4.4) emerged from measurement rather than from design, and it multiplies both qubit count and circuit depth. This mirrors the AES resource-estimation literature [14, 17], where the reversible cipher implementation, not the $2^{n/2}$ query count, dominates the cost.

Scope. Grover is not the only quantum threat to symmetric cryptography, and possibly not the most interesting one. Superposition attacks based on Simon’s algorithm give *exponential* advantage against certain constructions [20, 21], a qualitatively different threat model that the quadratic-speedup framing does not capture.

8 Limitations

1. **A 4-bit key is not cryptography.** It is a controlled testbed for oracle construction, amplification, noise and mitigation. Nothing here bears on the security of any deployed cipher.
2. **Gate counts are upper bounds.** The S-box is exact but synthesised from its permutation matrix by quantum Shannon decomposition [30], costing roughly 95 CX. A hand-optimised Toffoli network [2, 3] would be substantially cheaper, so all reported counts are upper bounds rather than optimal estimates.
3. **The noise model is a simplification.** Uniform depolarizing and readout error only, with no T_1/T_2 relaxation, crosstalk, leakage, parameter drift or correlated readout error. Real devices are both worse and less uniform.

4. **Noiseless rz is an idealisation, and a load-bearing one.** Virtual Z gates have zero duration [25] and IBM reports no error for them, but the measured error is small rather than zero. Since **rz** is the most common gate after transpilation (roughly half of all gates), assigning it even 10^{-4} error would contribute comparably to the CX channel. Our noise estimates are therefore optimistic in this specific respect, which strengthens rather than weakens the negative result of Section 6.3.
5. **Mitigation is not correction.** Post-processing a classical histogram corrects no quantum state; Experiment C measures precisely where that boundary lies.
6. **Calibration is exponential** in the number of measured qubits (2^n circuits). Tractable here only because the key register alone is measured; beyond roughly ten measured qubits a tensored or subspace-reduced method [27] is required.
7. **M is obtained by classical brute force** in order to set k^* . This is legitimate for studying algorithmic behaviour, but a genuine attacker ignorant of M would use the exponential-search schedule of Boyer et al. [8].
8. **Energy claims are deliberately weak.** Trajectory sampling reduces memory from 4^n to 2^n , and runtime maps loosely onto energy consumed [23]; reversible computation carries no thermodynamic lower bound per operation [4, 22]. We make an algorithmic choice with a favourable scaling consequence and do not claim to have measured energy savings.

9 Conclusions and future work

We built and exhaustively validated a reversible quantum oracle for known-plaintext key search against a toy SPN, confirmed that Grover amplification matches theory to 4.33×10^{-15} , and showed that at contemporary error rates the attack is not degraded but eliminated, with output entropy 3.995 of 4.000 bits. Classical readout-error mitigation recovers a readout-limited signal almost completely ($0.8408 \rightarrow 0.9652$, ideal 0.9613) and a gate-error-limited one not at all ($0.0574 \rightarrow 0.0596$). Recovery requires two-qubit error rates at or below 3.0×10^{-3} , and the quadratic query saving is bought with circuits growing to 33,360 CX gates by $n = 8$.

Four extensions follow directly:

- **Zero-noise extrapolation** for gate error [24, 31]. The noise-scaling machinery Experiment D already uses is exactly the mechanism ZNE requires.
- **Hand-optimised Toffoli S-box synthesis**, converting the resource counts from upper bounds into tight estimates and making deeper configurations simulable.
- **Scalable calibration** [27] to pass the 2^n calibration wall and reach wider key registers.
- **Superposition attacks** in the Simon’s-algorithm family [20, 21], which threaten certain symmetric constructions exponentially rather than quadratically.

Reproducibility. All code, seeded configurations, raw JSON records including full output distributions, and figure-generation scripts are released with this report. Every number printed above is generated directly from the saved records, so the report cannot drift from its data.

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